

DULUTH INTL, MN, USA

WMO: 727450

Lat: 46.837N

Lon: 92.183W

Elev: 1433

StdP: 13.95

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14913

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.3	-12.1	-24.8	1.5	-16.3	-20.4	1.9	-10.7	25.9	17.0	24.1	14.4	10.4	310	0.673

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	84.4	69.7	81.4	67.5	78.6	65.7	72.5	81.2	70.2	78.4	68.1	75.6	11.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.4	114.1	77.8	67.2	105.5	75.5	65.1	98.1	72.9	37.1	80.7	35.0	78.3	33.2	75.6	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.9	21.6	19.3	-23.3	90.0	5.4	2.7	-27.2	91.9	-30.3	93.5	-33.4	95.1	-37.3	97.0		
			-23.4	75.9	4.7	2.2	-26.8	77.4	-29.6	78.7	-32.2	79.9	-35.7	81.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.5	10.8	14.9	26.7	39.2	51.4	60.9	66.9	65.2	57.4	44.4	30.1	16.9
	DBStd	21.70	13.35	12.38	11.84	9.01	7.97	6.85	5.98	5.64	7.88	8.41	10.62	12.02
	HDD50	5212	1214	984	726	339	78	5	0	0	22	217	600	1027
	HDD65	9173	1679	1404	1188	774	429	157	50	66	247	639	1048	1492
	CDD50	1754	0	0	2	15	120	332	523	471	245	44	2	0
	CDD65	242	0	0	0	0	6	33	108	73	21	1	0	0
	CDH74	1925	0	0	0	3	101	315	868	497	134	7	0	0
	CDH80	406	0	0	0	0	20	62	207	100	17	0	0	0
Wind	WSAvg	9.9	10.5	10.5	10.2	11.0	10.4	9.2	8.5	8.4	9.3	10.4	10.4	10.3
Precipitation	PrecAvg	30.6	1.1	0.9	1.7	2.4	3.3	4.1	3.7	3.9	3.6	2.7	1.9	1.3
	PrecMax	43.4	4.7	2.7	5.1	8.2	7.7	10.0	8.7	10.3	9.4	6.8	5.4	3.7
	PrecMin	19.8	0.1	0.1	0.2	0.2	0.2	0.6	0.8	0.7	0.8	0.5	0.2	0.1
	PrecStd	4.9	0.9	0.6	1.1	1.4	1.5	1.7	1.7	2.0	1.9	1.6	1.3	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.7	44.9	60.8	70.9	82.7	86.4	89.1	87.3	82.3	73.9	60.8	41.9
		MCWB	35.4	38.9	52.6	62.7	71.4	73.6	72.3	68.3	59.3	49.9	36.5	
	2%	DB	35.3	39.2	52.3	64.4	76.4	81.1	85.1	82.7	78.0	66.9	53.4	37.7
		MCWB	33.0	35.4	44.3	49.6	59.0	66.4	70.5	69.2	66.2	56.3	47.7	34.8
	5%	DB	32.0	35.4	46.4	58.8	71.0	77.6	82.1	79.3	73.9	61.7	48.2	34.8
		MCWB	30.4	32.3	40.1	46.8	56.4	64.1	68.5	66.9	63.9	53.1	42.4	32.7
	10%	DB	29.0	32.0	41.3	53.8	66.0	74.1	79.1	76.3	70.2	57.4	44.3	32.6
		MCWB	27.3	29.6	36.7	43.5	54.0	62.1	66.7	65.3	61.9	50.6	40.1	31.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.3	39.7	53.9	56.8	66.3	73.9	75.8	75.1	71.2	62.7	53.2	38.7
		MCDB	39.8	43.5	58.1	64.9	77.1	82.7	85.8	82.0	79.2	70.6	57.3	39.9
	2%	WB	33.2	35.9	45.4	51.0	61.9	69.2	73.0	71.8	68.4	58.4	47.7	35.1
		MCDB	35.1	38.5	51.0	62.1	71.4	78.5	82.1	80.1	74.8	64.0	52.9	37.1
	5%	WB	30.4	32.7	40.6	47.5	58.6	66.3	70.6	69.2	65.9	54.8	43.6	33.1
		MCDB	32.0	35.1	45.6	58.1	68.1	73.6	78.9	76.2	71.9	60.9	47.2	34.8
	10%	WB	27.3	29.7	37.0	44.4	55.4	63.8	68.4	67.0	63.1	51.2	40.3	31.0
		MCDB	28.9	32.1	41.3	52.8	64.8	71.7	76.4	73.9	68.5	56.3	44.0	32.8
Mean Daily Temperature Range	5% DB	MDBR	15.5	16.6	16.1	17.0	19.5	19.2	19.0	17.9	17.2	14.7	12.8	13.0
		MCDBR	14.4	17.1	20.1	26.2	27.4	24.3	22.3	21.2	21.2	21.6	18.3	13.6
		MCWBR	12.4	13.4	13.4	15.6	14.4	12.9	11.3	10.7	11.8	13.1	12.6	11.2
	5% WB	MCDBR	14.0	14.4	18.9	24.5	23.9	21.6	20.6	18.9	18.3	19.2	17.4	12.3
		MCWBR	12.3	11.6	13.2	15.3	14.4	13.2	12.3	11.2	11.7	13.0	13.0	10.6
Clear-Sky Solar Irradiance	taub		0.241	0.259	0.303	0.341	0.368	0.366	0.374	0.376	0.353	0.307	0.282	0.259
	taud		2.412	2.375	2.377	2.392	2.366	2.398	2.395	2.415	2.452	2.572	2.528	2.427
	Ebn at Noon		279	294	293	290	285	285	281	276	273	274	260	256
	Edh at Noon		21	27	32	35	37	36	36	34	30	23	19	19
All-Sky Solar Radiation	RadAvg		461	807	1173	1464	1687	1825	1939	1626	1222	742	442	328
	RadStd		50	74	110	147	125	128	79	119	86	95	51	33

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126	+54
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page